

Introduction to Programming, 7.5 credits

LIS020

Course description

Autumn 2026

Version 1 (15 July)

Teachers

Maria Skeppstedt
Mail: maria.skeppstedt@ling.su.se
Office: C236

Nikolaj Lindberg
Mail: nikolaj.lindberg@ling.su.se

Amanda Kann
Mail: amanda.kann@su.se
Office: C268

Course content

This course provides knowledge and skills in solving simple language and speech technology problems by constructing programs in a contemporary programming language.

Learning outcomes

In order to pass the course, students are expected to be able to:

- solve simple language and speech technology problems by constructing programs in a contemporary programming language,
- apply and describe the rules of good programming style, including documentation, error handling, structuring, and flexibility,
- understand, troubleshoot and modify specific programs,
- use modern tools to write and handle programming code,
- use external libraries for contemporary language and speech technology applications.

Required reading

The main course book is:

- Charles Severance. 2024. *Python for Everybody*. Available online: <https://www.py4e.com/>

In addition, we will make extensive use of parts of the following:

- The Python Standard Library. Available online: <https://docs.python.org/3/library/index.html>
- The Python Tutorial. Available online: <https://docs.python.org/3/tutorial/index.html>

Additional reading of at most 100 pages may be handed out by the teachers during the course.

Education

Instruction is given in the form of lectures and laboratory sessions.
Instruction is given in English, on campus.

Forms of examination

The examination of this course is carried out as specified below. Note that there will be an oral exam for Lab assignment 3. This means it is mandatory to attend, either Wednesday October 21 or Monday October 26.

- Lab assignment 1.
Regular deadline: Monday, October 5.
(Re-examination deadline: Monday, October 19.)
Hand in through Canvas
- Lab assignment 2.
Regular deadline: Wednesday, October 14.
(Re-examination deadline: Friday, October 31.)
Hand in through Canvas
- Lab assignment 3.
Regular deadline: Wednesday, October 21.
(Re-examination deadline: Monday, November 2.)
Hand in through Canvas, on deadline.
Oral presentation either Wednesday October 21
or Monday October 26
- Lab assignment 4.
Regular deadline: Wednesday, October 28.
(Re-examination deadline: Monday, November 9.)
Hand in through Canvas
- Written exam.
Regular exam: Monday October 12
Re-take exam: Tuesday November 10 (Only for those who did not pass the regular exam.)

General Schedule

The basic structure of a day in this course is that both group 1 and group 2 start at 10:00 in the computer rooms (Språkstudion, rooms may vary but this is specified in TimeEdit). The lecturer will move between the rooms, being physically present in one room and on video link in the other room. This time will mainly be spent lecturing and group discussions. Group 1 will then have a session for individual work at 13:00-14:45, and group 2 at 15:00-16:45. During this time the computer labs are reserved for this course, and teachers will generally be available during the first hour. Due to computer lab booking constraints, there will be some deviations from this basic structure, so make sure to check the TimeEdit schedule.

Course overview

See separate document for course overview.

Grading criteria

The criteria below have been accepted by the Department of Linguistics board at the meeting 2024-05-14 och are valid from autumn 2024.

The examination consists of a written exam which is graded pass/fail (G/U), and four individual lab assignments which are graded pass with distinction/pass/fail (VG/G/U). Criteria for each respective grade are found below. For a passing grade, all criteria at the pass level must be satisfied, and for passing with distinction at least half (rounded up) of the pass with distinction criteria must be satisfied.

Principles for computing the final grade

Final grade (A-F)

- For a passing grade (A-E) you are required to pass the written exam and all individual assignments.
- The passing grade (A-E) is computed based on the grades of the individual lab assignments in the following way, based on the number of assignments graded pass with distinction (VG):
 - 0: E
 - 1: D
 - 2: C
 - 3: B
 - 4: A

Expected study results	Pass	Pass with distinction
solve simpler language and speech technology problems by constructing programs in a modern programming language,	– Design programs to solve simple language and speech technology problems (L/T)	– Design more comprehensive programs to solve language and speech technology problems (L)
apply and explain rules of good programming style, including documentation, error handling, structuring and flexibility,	– Document code with at least one common type of documentation (L) – Follow general principles for naming variables, functions and classes (L)	– Document code with a reasonable number of clear inline comments, docstrings and help texts (L) – Consistently use informative and conventional names for variables, functions and classes (L) – Consistently use a flexible and expedient code structure (L)
understand , debug and modify given programs	– Extend existing programs with new functionality, given references to relevant parts of the program (L) – Identify common errors in code (T) – Derive the function of a given piece of code (T)	– Independently expand existing programs with new functionality (L)
use external libraries for modern language and speech technology applications	– Use external libraries for language and speech technology applications, given references to appropriate libraries and functions (L)	– Use external libraries for language and speech technology applications, by independently reading existing documentation (L)
use modern tools to write and manage program code	– Use automatic tools for analysis and/or generation of code (L)	– Use tools for version management (L)

The matrix indicates whether the criteria are primarily examined through (L)ab assignments and/or the written exam (T). This information is to support course development and does not imply any promises about the design of the examination.

Additional information

Special pedagogical support

Stockholm University provides special pedagogical support to students with documented long-term disabilities. For more information on how to apply for pedagogical support, please see:

<https://www.su.se/english/education/during-your-studies/studying-with-a-disability>

The following is required of you:

- Please contact the Coordinator for special pedagogical support for special pedagogical support as soon as possible upon your arrival in Stockholm in order to discuss your individual needs and apply for special pedagogical support.
- Well in advance of the examination – a minimum of 3 weeks – please inform the department coordinator/examiner if you need an alternative/accommodated form of examination.

Please note that the coordinator for special pedagogical support provides recommendations, but it is always the examiner who decides whether these recommendations can be met given the requirements of the course plan.

Department coordinator for special pedagogical support:

- Åsa Gustafsson
- Phone: 08-16 23 37

Stockholm University rules and regulations

For the university's rules in regard to education, examination, written exams and rules on disciplinary actions and reprimands according to the Higher Education Legislation, we refer to URL:

www.su.se/styrdokument

A selection of the University's applicable rules and regulations for Education are available in English: <https://www.su.se/english/staff/organisation-governance/governing-documents-rules-and-regulations>

(NOTE: what follows is a standard text that we need to show you, but during the introduction lecture we will be more specific about what all this means for this particular course)

Disciplinary actions

From the Guidelines for disciplinary actions in the Stockholm University Governing Documents – Rules & Regulations, Book 2:

Disciplinary actions are regulated by chapter 10 in the Higher Education Legislation (1993:100) and can be enforced for students who:

- Attempt to cheat at examination or any study achievement to be marked by using unsolicited measures
- Disrupt or hinder teaching, examination or other activities within higher education
- Disrupt activity at the university's library or at another higher education facility
- Expose another student or employee at university for harassment or sexual harassment as defined in chapter 1 §4 Discrimination Law (2008: 567).

Disciplinary actions must not be enforced later than two years after the misconduct has been committed. Certain forms of disciplinary misconduct, e.g., disruptive behavior (threat or violence), harassment (general, sexual, or hate crime), document forgery or untrue certification, can also fall under criminal law and therefore be prosecuted.

Responsible use of AI tools

The four lab assignments must be conducted individually, and without the use of AI-supported code generation tools. The students are expected to be able to explain their code to the teachers.

Plagiarism and cheating

Plagiarism is defined as imitating or copying someone else's text or idea without referring to where they originated. All submitted assignments are to be written with your own words and to be built on your own analysis and your own thoughts on the topic. To express your ideas with your own words is part of the learning process and improves the ability to think independently.

When you report others' texts or ideas, or your own texts and ideas that you have earlier submitted as an assignment or published, it must be clear for the reader that you are citing or referring to another text and from where you have taken the information. All citations must be correctly marked with citation marks (or in block citation if they are longer excerpts) and source denotation. Any text must not mostly consist of citations. These should only be used to illustrate your own reasoning. Remember to use your own words even in oral presentations. All work that you quote or refer to must be listed.

- For books, list: author(s), year, title (in italics), publishing location, publishing house and pages.
- For articles in journals, list: author(s), year, title, journal name and volume number (in italics), issue number and pages.
- For articles published in proceedings of symposia and conferences or collection volumes, list: author(s), year, title, editor name (s) followed by (ed.) in brackets, Proceedings/volume title (in italics), publishing location, publishing house and pages.
- For internet material that cannot be sorted in any of the categories above, list: author(s), title, internet address, download date.



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If you are uncertain about what counts as plagiarism, ask your teacher. Suspected plagiarism cases are passed on to the university's Disciplinary Board and can lead to suspension. For more information, please read the section on examination in Stockholm University's governing documents.